Numerical Analysis Assignment 6

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Question 1

No! Let $P_6(x)$ and $P_5(x)$ denote the 6degree polynomial and 5 degree polynomial respectively. To find the interceptions of these two polynomials is to find roots of $q(x) = P_6(x) - P_5(x)$, the $q(x)$ is a 6 degree or less polynomial. By the fundamental theorem of algebra, there are at most 6 interceptions of these two polynomials.

Question 2

(a) No 1 degree polynomial passes through the points.

(b) There is exactly one 2 degree polynomial that passes through these three points. It is

$$p(x) = -\frac{4}{3}x^2 + \frac{11}{3}x + 1.$$

(c) There are infinitely many polynomials that pass through the three points. Possible one is

$$p(x) = -\frac{4x^4}{3} + 13x^3 - 38x^2 + \frac{89x}{3} + 10$$

Question 3

By Lagrange interpolation,

$$\begin{aligned} L_1(x) =& \frac{(x-2)(x-4)}{(0-2)(0-4)} = \frac{(x-2)(x-4)}{8} \\ L_2(x) =& \frac{(x-0)(x-4)}{(2-0)(2-4)} = -\frac{x(x-2)}{4} \\ L_3(x) =& \frac{(x-0)(x-2)}{(4-0)(4-2)} = \frac{x(x-2)}{8} \end{aligned}$$

Then,

$$y(x) = -2L_1(x) + L_2(x) + 4L_3(x) = \frac{3x}{2} - 2.$$

Question 4

By Lagrange interpolation, if n points are interpolated, the coefficient of the $n-1$ degree item is

$$c_{n-1} = \sum_{i=1}^{n} \left(y_i \prod_{\substack{j=1 \\ j \neq i}} (x_i - x_j)^{-1} \right).$$

Then, the coefficient of x^3 in $P^3(x)$ is

$$c_3 = -1 \times 0 + y \times \frac{8}{3} + 3 \times (-2) + 2 \times \frac{1}{3} = 6,$$

$$\frac{8y}{3} - \frac{16}{3} = 6,$$

then, $y = 17/4$.

Question 5

For disjoint unequal $x_1, x_2 \ldots, x_m$, The *Vandermonde* matrix is

$$A_m = \begin{bmatrix} 1 & x_1 & \cdots & x_1^{m-1} \\ 1 & x_2 & \cdots & x_2^{m-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_m & \cdots & x_m^{m-1} \end{bmatrix}$$

By applying column and row applications to the matrix,

$$A_m \xrightarrow[\text{for } i=2,\cdots,m]{C_i - x_1 C_{i-1} \to C_m} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & x_2 - x_1 & \cdots & x_2^{m-2}(x_2 - x_1) \\ \vdots & \vdots & & \vdots \\ 1 & x_m - x_1 & \cdots & x_m^{m-2}(x_m - x_1) \end{bmatrix} \xrightarrow[\text{for } i=2,\cdots,m]{(x_i - x_1)c_i \to c_i} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{1} & A_{m-1} \end{bmatrix}.$$

Then,

$$\det(A_m) = \det(A_{m-1}) \cdot \prod_{k=2}^{m} (x_k - x_1).$$

Since $\begin{vmatrix} 1 & x_{m-1} \\ 1 & x_m \end{vmatrix} = (x_m - x_{m-1}) \cdot 1$, then

$$\det(A_n) = \prod_{i=1}^{m-1} \prod_{j=i+1}^{m} (x_j - x_i) = \prod_{1 \le i \le j \le m} (x_j - x_i).$$

Since $x_1, x_2 \ldots, x_m$ are disjoint unequal, so the *Vandermonde* determinate $\det(A_m) \neq 0$. So, *Vandermonde* matrix is nonsingular. Thus if the interpolation problem has a solution, it must be unique.